Estimation of Unobservable Household Time-Use Decision-Making Parameters

By: Ari Alexander Etin

Honors Thesis
Economics Department
The University of North Carolina at Chapel Hill

Thesis Advisor: Prof. Luca Flabbi, PhD
Faculty Advisor: Klara Peter, PhD

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Abstract

Effective policy analysis is predicated upon quantitative models, and forward-looking simulation is often limited to extensions of past events. However, the method of simulated moments (MSM) allows for estimates of unobservable parameters, making possible truly forward-looking quantitative models. In labor economics, existing estimates of household time-use decision parameters, performed by del Boca and Flinn using 2005 PSID data, are outdated by two decades and contain few robustness checks. Using the MSM, we estimate preference and efficiency parameters under a collaborative behavioral model. Data are a subset of the 2005, 2017, 2019, 2021, and 2023 PSID data, filtered by similar requirements to del Boca and Flinn. We successfully replicate prior results and show mixed robustness to sample and initial search conditions. Moreover, updated decision-making parameters suggest significant differences from 2005. This research begets further questions of the evolving nature of household time use decision dynamics and improves on existing parameter estimates for policy simulation.

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1 Introduction

Labor and household economics have made substantial theoretical progress through the consideration of individuals as making decisions in the context of a domestic unit, not independent of others. In doing so, however, policies intended to target household decision-making have treated households as a homogenous unit. Unfortunately, research suggests that taxation, family work policies, child incentives, divorce law, and other such policies require the consideration of within-household factors (Chiappori and Mazzocco, 2017; Lazarus and Mandel, 2023). In the absence of validated estimates of unobservable parameters, quantitative models cannot be developed to predict the effects of proposed policy shocks.

Early estimates of preference and Cobb-Douglas efficiency parameters of households were presented by del Boca and Flinn in 2012, using 2005 PSID data. This thesis extends their estimates of unobservable household decision-making parameters, using the same collaborative model and estimation strategy. Addressing the two-decade gap in prior data and the present, this research first replicates the 2012 results and then extends the estimation strategy to 2017–2023 PSID samples.

Actual household resource allocation and consumption decisions are well-studied. Prior research has explored household decisions regarding time use (Hofmarcher and Plug, 2022; Lazarus and Mandel, 2023; Lise and Yamada, 2019; Waddell et al., 2021; Yang, 2025), spending (Chiappori et al., 2024; Lechene et al., 2022; Lise and Seitz, 2011; Lise and Yamada, 2019), and income (Ahmed et al., 2011; Bertrand et al., 2015). These observable characteristics provide valuable information about what is happening under present circumstances but shed little light on why present conditions have the effects they do. Therefore, policy analyses are constrained to shocks already observed.

This paper therefore extends prior research by looking not directly at time use, but the preferences of individuals within contemporary households. Del Boca and Flinn developed models of household behavior under non-collaborative, collaborative, constrained collaborative, and mixed behavioral regimes. The models include unobserved parameters which reflect the importance of each partner's utility in the household's time-use decision. Crucially, del Boca and Flinn (2012) use the method of simulated moments (MSM) for its unique efficacy in estimating these parameters. Because their focus was not on the method itself, del Boca and Flinn do not report certain key implementation details, including the initial conditions for their optimization, which algorithms were used, and the specifics of their bootstrapping process. The aims of this research were to 1) replicate del Boca and Flinn (2012)'s analysis, 2) test the robustness to sample selection, 3) test the robustness to initial conditions, and 4) explore recent samples of PSID data, using the MSM.

Replication is largely successful, with further tests showing robustness to sample, mixed robustness to initial conditions, and differences in unobserved parameters between 2005 and more recent years. The following sections will provide (2) an overview of past research, (3) a discussion of the dataset, (4) a theoretical model, (5) an empirical model, (6) empirical results, (7) a discussion, and (8) a conclusion.

2 Literature Review

As established, expansion beyond the unitary model of household decision-making is warranted and necessitated by this research. The simplest alternative, from which many models in this domain expand upon, involves a household with two decision makers, who each benefit from their own private consumption of a resource (time or goods) and some externality, such as a public good. Alternatives to unitary decision-maker models vary among three domains: assumptions regarding cooperation (e.g., Del Boca and Flinn, 2012), number of public goods (e.g., Chiappori et al., 2024), and dynamism (e.g., Chiappori and Mazzocco, 2017).

2.1 Cooperative or Non-Cooperative

Borrowing insights from game theory, households like these can be thought of as two-player games. As a game with two players, the simplest model of behavior is a non-cooperative game, where each player acts according to their best response function, ultimately finding a Nash equilibrium. Indeed, some papers do include this approach (Del Boca and Flinn, 2012; Del Boca and Flinn, 2014). In doing so, they in effect violate the assumption that the household makes collective decisions, and therefore these papers also include cooperative models, where households maximize a joint utility function, arriving at a pareto frontier.

This collaborative approach aligns with popular notions of marriage, as an arrangement of joint decision making and combined resources that can produce better outcomes for each member than they might independently achieve. As such, Pareto optimal (collaborative) models of household behavior are ubiquitous (Chavas et al., 2018; Del Boca and Flinn, 2012; Lechene et al., 2022; Lise and Seitz, 2011). In effect, these models presume that the household maximizes a joint utility function (a weighted average of individual utility functions), with some resources afforded to each individual and some shared good(s).Compared to such a model, the conceptual advantage of the non-cooperative model is that it does not require individuals to entirely suppress their interests for the 'greater good' and, indeed, in at least one direct comparison, Nash equilibria outperform an unconstrained[1] Pareto optimal (coop-

[1]Del Boca and Flinn, 2012 compare four models of behavior: Nash equilibria, Pareto optimality, constrained Pareto optimality, and endogenous household interaction. Nash equilibria only outperform (in terms

erative) model of household behavior (Del Boca and Flinn, 2012). However, del Boca and Flinn (2012; 2014) find that, once constraints are added to the possible solution set, Pareto optimal models of household behavior describe observed behavior better than the Nash equilibria. In particular, constrained Pareto optimal regimes limit choices to the section of the pareto frontier not worse for either party than the Nash equilibrium. Del Boca and Flinn (2012; 2014) call this model constrained Pareto optimality.

2.2 Additional Considerations: Intertemporal Models and Multiple Public Goods

Marriage may last several years, even a lifetime (if death does do them part). Allowing decisions to evolve with preferences and material conditions in a relationship has been afforded by multi-period (panel or semi-panel) data. Unfortunately, intertemporal models are scarcer and less developed than static ones (Chiappori and Mazzocco, 2017). However, one recent study by Lise and Yamada suggests that adding heterogeneity over time is relevant for large shocks, but that household allocative decisions are robust to smaller ones (Lise and Yamada, 2019). Therefore, the dearth of literature on intertemporal models may stem from the assumption that incorporating the small shock of time may over-complicate models not explicitly exploring other, larger shocks. Insofar as intertemporal considerations are discussed, it is to allow switching between behavioral strategies (Del Boca and Flinn, 2012). For instance, del Boca and Flinn (2012) innovated on constrained cooperative models by allowing a household to choose whether to act cooperatively. This required an identification of a theoretical 'discount factor' but still empirically focused only on one period.

Up until this point in the review, household models have assumed one public good: partners are not allowed to differentially prefer one type of good over another (e.g., a clean kitchen versus home-cooked meals). Assuming that each partner might have their own preferred category of goods, one solution is to create one good for each partner. Mathematically, this is simple enough; however, operationalizing the categories is complex, requiring many assumptions about the preferences of potentially heterogeneous populations. The contents of each category must align with the differential preferences of each individual and consistently map to one partner or the other across many households. Moreover, data providing a sufficient degree of specificity on resource allocations are scarce. Despite the limitations,

of explanatory power) the unconstrained Pareto optimality condition. Presumably this is because individuals would prefer not to form a collaborative household if they would be better off not collaborating. constrained Pareto optimality, where the choices are limited to the section of the pareto frontier not worse for either party than the Nash equilibrium, outperforms Nash equilibria. endogenous household interaction outperforms all other models and allows households to switch between Nash equilibria (noncooperative) and constrained Pareto optimality (cooperative) regimes

Chiappori et al. (2024) apply this technique to childcare-related goods and non-childcare expenses using a uniquely detailed dataset on Japanese household consumption by member. They find that this heterogeneity in member's preferences for household goods is reflected in the data, and that husbands prefer childcare expenditures less than wives.

2.3 The Choice of Behavioral Regime in this Paper

As most advanced models incorporate some degree of collaboration between household members, we elected to do the same. However, to minimize complexity and establish basic viability of the project, the unconstrained Pareto optimal behavioral model was chosen. Having made these choices, future work in this domain is well-equipped to expand on the present approach with more nuanced models.

3 Data

Data for this project are derived from 2005, 2017, 2019, 2021, and 2023 Panel Study of Income Dynamics (PSID) samples. The PSID, as the name suggests, is a panel survey of 5,000 families and their descendants, first selected as a nationally representative sample in 1968. The PSID is known for its high degree of data quality, documentation, and availability. As a result, it has become a staple of economic research and is the dataset used by del Boca and Flinn (2012). Therefore, use of this dataset allows for more direct comparisons of results to prior literature.

A core limitation of PSID data is the inconsistency of questions from year to year. Although this allows the survey to adapt to emerging trends or shifting populations, it also adds potential error to explanatory variables. For instance, older versions of the PSID use the husband as the reference person in a household. In 2015, a landmark supreme court decision legalized same-sex marriage in the US, and subsequent panels of data (2017 onwards) included substantially more same-sex households than prior years. To accommodate this, the PSID has begun to ask for a 'head of household' or 'reference person.' Note that 'husband,' head of household,' and 'reference person' do not necessarily align—indeed, the definition of a reference person can misguide interpretation of results.

3.1 Exclusion Criterion: Same-Sex Households

The sample criteria used differ from those of del Boca and Flinn (2012) in that same-sex households were explicitly excluded. Men and women in same-sex households do not by definition have different preferences and time-use efficiencies. Moreover, same-sex households

are often underrepresented in research. Therefore, this subsection explores the decision to exclude same-sex households from the analysis.

The method of simulated moments uses observed time use (and income) to 'reverse-engineer' the underlying distributions of household time-use decision variables. Therefore, this discussion primarily involves a comparison of same-sex and different-sex households' time use. Due to same-sex marriage being legalized in 2015, data from the 2005 PSID sample identify only one same-sex household—two individuals out of over 12,000 (before cleaning) observations. Samples from 2017 onward are subsequently used.

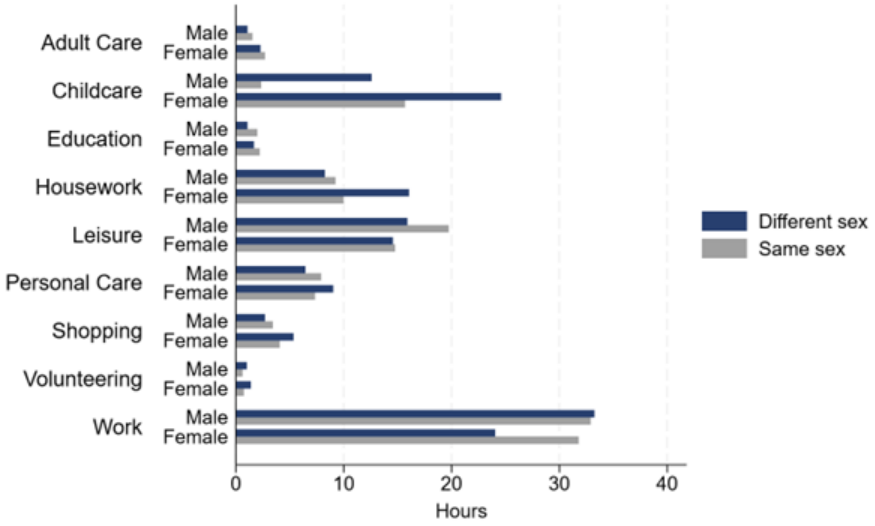
PSID respondents were directly asked about their typical weekly time use along nine dimensions: Adult Care (assisting another adult with daily activities), Childcare, Education, Housework, Leisure, Personal Care, Shopping, Volunteering, and Work. The exact questions are included in Appendix B. These questions (or the equivalent for other years) are the ones typically used in PSID analyses, including by del Boca and Flinn (2012). Table 1 provides summary statistics for these variables, comparing same-sex and different-sex households. There appear to be significant differences in time use on housework, childcare, volunteering, education, leisure, and work when comparing an average individual in a same-sex household to one in a different sex household. Further segmenting the data, Figure 1 explores time use by sex and household type. Same sex male and female households show incredibly similar time use along most dimensions except childcare (also noted in prior literature, e.g., Bauer, 2016) and leisure (which may also be related to childcare or the lack thereof).

Table 1: Time Use by Activity and Household Type

	Different Sex	Same Sex
N	33,359	495
Time in housework ***	12.388 (11.992)	9.869 (8.185)
Time in personal care	7.877 (6.533)	7.657 (7.211)
Time in shopping	4.141 (5.036)	4.006 (5.071)
Time in child care ***	19.148 (36.572)	12.404 (30.052)
Time in adult care	1.683 (11.260)	2.483 (14.814)
Time in volunteering ***	1.212 (3.588)	0.723 (2.468)
Time in education ***	1.384 (5.027)	2.101 (6.726)
Time in leisure ***	15.152 (13.077)	16.677 (16.597)
Time spent working ***	28.749 (21.275)	32.826 (20.272)

Observed time use of members in same-sex and different-sex households. Data from years 2017–2023. Standard deviations in parentheses. Despite slight sample variation (due to incomplete time use data) demographic information aligns closely with Table 2. Significance scores based on p-value for testing the equality between groups using t-test for continuous variables. Key: * = 90% confidence, ** = 95% confidence, *** = 99% confidence.

Figure 1: Weekly Time use by Household Type, Sex, and Activity



Mean weekly time use in hours, with observations grouped by sex and household type.

Turning now to a demographic comparison of same-sex and different-sex households, Table 2 indicates that the same-sex sample significantly differs from the different-sex sample in several key dimensions: age, race, education, family size, family income, and employment status. There are no significant differences in the likelihood of a household being below the poverty line (adjusted for household size but not age). This is understandable if family size is correlated with family income (the decision to have children being a financial one).

Furthermore, intra-household characteristics appear to differ between same-sex and different-sex households. For within household demographic differences by household type, see Table 3. To address the challenge of identifying a non-random assignment to 'primary' and 'secondary' partner, mean absolute differences are reported. This was found to reveal within-household disparities which would have been masked, had the 'reference person' of the household been used. For instance, age gaps in same-sex households were nearly zero with the 'reference person' method but shot up to over five years when taking mean absolute differences.

Comparing within-household differences, a more vivid depiction of same sex households emerges. In this sample, they tend to have larger age gaps, are more likely to be interracial, and have smaller differences in labor market outcomes (earnings, current employment status, and prior year employment status). These findings (reported in Table 3) support economic reasoning: if a gender wage gap exists, then households where individuals have the same gender should have smaller labor income disparities. Moreover, if the population of people willing to enter into a same-sex household is smaller than that of people willing to enter into

Table 2: Demographic Characteristics by Household Type, 2017–2023

	Different sex	Same sex
N	37,061	550
Age***	47.116 (15.121)	39.911 (13.248)
Years of completed education, imputed***	13.880 (2.576)	14.538 (2.157)
Family unit size***	3.270 (1.385)	2.487 (0.831)
Prior year total family income***	122.768 (135.081)	104.706 (86.603)
Prior year labor earnings	48.227 (80.066)	45.308 (54.792)
Race***		
White	0.678	0.636
Black	0.235	0.271
Other	0.082	0.067
Unknown	0.005	0.025
College degree***		
No	0.472	0.424
Yes	0.412	0.495
Unknown	0.115	0.082
Currently working***	0.707	0.776
Full-time status last year***		
No work (0-49)	0.248	0.178
Part-time (50-1749 hours)	0.197	0.191
Full-time (1750+ hours)	0.555	0.631
Low-income family last year	0.054	0.069

Comparison of demographic characteristics for same-sex and different-sex households. Data are combined for all years and do not reflect the subsequent data cleaning steps. Standard deviations in parentheses for continuous variables. Significance scores based on p-value for testing the equality between groups using t-test for continuous variables and Pearson χ^2 tests for factor variables. Total family income, labor earnings and labor earnings are in $1000's, all prices in 2005 USD. Key: * = 90% confidence, ** = 95% confidence, *** = 99% confidence.

a different sex household, then assortative mating might be weaker. However, there appears to be no significant difference in education gaps within same and different sex households (whether measured years of education or having gone to college), suggesting that assortative mating is not exclusive to different sex households.

3.2 Data for the Present Analysis

Having excluded same-sex households, the remaining criteria were intended to best replicate the data in Del Boca and Flinn (2012). Observations were limited to married different-sex couples who were both employed the previous year, had positive total weekly income, did not make more than $1000/week in non-labor income, and spent no more than 100

Table 3: Differences in Demographic Variables

	Different sex	Same sex
N	37,061	550
Absolute Diff—Age***	3.549 (3.714)	5.166 (5.773)
Absolute Diff—Years of completed education, imputed	1.542 (1.670)	1.452 (1.597)
Absolute Diff—Prior Year Labor Earnings **	46.567 (94.027)	35.379 (60.033)
N	18,520	278
Same race***	0.903 (0.296)	0.824 (0.382)
Same college	0.646 (0.478)	0.647 (0.479)
Same current employment status***	0.701 (0.458)	0.798 (0.402)
Same Full-time status last year***	0.510 (0.500)	0.619 (0.487)

Within-household differences in categorical and demographic variables. First set of variables are mean absolute differences. Second set of variables are the probability that partners are of the same category. Standard deviations in parentheses for continuous variables. Significance scores based on p-value for testing the equality between groups using t-test for continuous variables. Labor earnings are in $1000's, all prices in 2021 USD. Key: * = 90% confidence, ** = 95% confidence, *** = 99% confidence.

Table 4: 2005 Observed variables, means (standard deviations)

	Present Analysis		Del Boca and Flinn, 2012	
Variable	Men	Women	Men	Women
h	43.589	33.159	45.706	38.588
	(12.110)	(13.498)	(8.546)	(10.512)
τ	7.994	15.763	7.787	14.920
	(6.741)	(10.557)	(6.418)	(9.428)
w	25.130	19.271	22.009	15.823
	(18.706)	(14.835)	(13.626)	(9.327)
Y	95.618		118.151	
	(62.061)		(182.526)	

Comparison of the observed means and standard deviations of time use and income variables. Two-sample independent t-tests confirm that there are no significant differences between the samples.

hours on work and housework (per person). Couples with incomplete information about both partners were dropped. Of the 2005 data, 972 couples remained—somewhat greater than the 823 used in Del Boca and Flinn (2012). Del Boca and Flinn (2012) do not report demographic characteristics other than time use and income, so we used a comparison of these—reported in Table 4—to confirm similarity of samples. No significant differences were observed. Demographic characteristics of our observations are reported in Table 5. Identical

selection criteria were used on the novel data, which produced larger samples for each year than 2005. Demographics of novel data are listed in Table 6.

Table 5: Summary Statistics by Gender, 2005

	Gender	
	Male	Female
N	972	972
Age***	40.831 (5.109)	39.435 (5.126)
Years of completed education, imputed***	13.579 (2.262)	13.898 (2.180)
Family unit size	3.818 (1.156)	3.818 (1.156)
Total family income last year	95.618 (62.061)	95.618 (62.061)
Prior year family labor earnings	89.620 (62.157)	89.620 (62.157)
Prior year labor earnings***	56.401 (46.051)	33.219 (29.763)
Prior year household weekly non-labor income	115.337 (189.855)	115.337 (189.855)
Prior year hourly wage***	25.130 (18.706)	19.271 (14.835)
Race***		
White	0.718	0.719
Black	0.224	0.218
Other	0.058	0.063
College degree***		
No	0.560	0.485
Yes	0.315	0.372
Unknown	0.126	0.143
Currently working***		
0	0.027	0.076
1	0.973	0.924
Full-time status last year***		
Part-time (50-1749 hours)	0.115	0.393
Full-time (1750+ hours)	0.885	0.607
Low-income family last year		
0	0.989	0.989
1	0.011	0.011

Demographics of the sample used for estimation. Standard deviations in parentheses for continuous variables. Significance scores based on p-value for testing the equality between groups using t-test for continuous variables and Pearson χ^2 tests for factor variables. Total family income, total labor earnings and individual labor earnings are in $1000's, all prices in 2005 USD. Key: * = 90% confidence, ** = 95% confidence, *** = 99% confidence.

4 Theoretical Model

Research that evaluates inequality of resource allocation within households makes the strong assumption that preferences are homogenous between partners. To illustrate the challenge with this approach, suppose a household where one member prefers an even split of tasks between individuals and the other strongly prefers to be solely responsible for household tasks. Were the former partner to dominate the bargaining, the demonstrated time allocation would appear equal and be interpreted as such. By contrast, a decision-making process that equally values each partner would result in an unequal allocation of time (though not as extreme as either partner would prefer). The first step in addressing this limitation is to model partner preferences.[2]

To do this, we select a behavioral model of household time-use decisions presented in del Boca and Flinn (2012). With the finding that most households collaborate—over 75% as per the del Boca and Flinn (2012) analysis—the pareto-optimal model is preferred for its simplicity.[3] The model presumes a household where the utilities of two individuals are considered in making a joint time-use decision—there may be more individuals, but their utilities are presumed to be part of a household product and therefore not directly accounted for in making the time allocation decision. Partners allocate time between household tasks, leisure, and wage-earning tasks.

The key technical assumption taken by del Boca and Flinn (2012) is that there are unique pareto-optimal actions $\left(\widehat{a_1^P}, \widehat{a_2^P}\right)$ for partner 1 and partner 2, given a particular weighted average of utility functions for each partner (u_1, u_2) (defined by a 'bargaining weight', (α), and set of available choices). As a collaborative model of household behavior, del Boca and Flinn (2012) specify that the household decides on courses of action to maximize household utility given by

$$W_i = \alpha_i u_{i,1}\left(a_{i,1}, a_{i,2}\right) \ + \ \left(1 - \alpha_i\right) u_{i,2}\left(a_{i,1}, a_{i,2}\right)$$

, with the constraints that the actions $(\widehat{a_1^P}, \widehat{a_2^P})$ exist in the set of available choices. Alpha, presumed homogenous and equal to 0.5 in the present paper, is the bargaining weight referenced at the beginning of this section.

Further specifying the model, individual utilities are similarly presumed to be weighted averages of utility from leisure (time that only benefits oneself) and a household good. The

[2]Subsequent steps involve allowing heterogeneity in the weight given to partners, discussed further in Section 7.

[3]Of the four behavioral models discussed in del Boca and Flinn (2012), pareto optimality was least evidenced; as discussed in limitations, further research should explore recent data with alternative behavioral models.

household good, in turn, is defined as a Cobb-Douglas product, a function of the time each partner directly spends on it and the total income (labor and non-labor) available:

$$
\begin{aligned}
u_{i,j} &= \lambda_j \ln l_{i,j} + (1 - \lambda_{i,j}) \ln K_i \\
K_i &= \tau_{i,1}^{\delta_{i,1}} \tau_{i,2}^{\delta_{i,2}} M^{1-\delta_{i,1}-\delta_{i,2}} \\
M_i &= w_{i,1} h_{i,1} + w_{i,2} h_{i,2} + Y_i
\end{aligned}
$$

with the following time-use constraint:

$$T = l_{i,j} + \tau_{i,j} + h_{i,j}$$

I now report on the gradients and optimal values of each partner's time use, as a function of exogenous variables. Because partner utility functions differ only in the actual values of coefficients, I represent equations for partner 1. For further simplicity and clarity, I omit the household subscript; however, the theoretical model allows for heterogeneity in all variables except T, the total amount of time available to each person in each week. Additionally, because leisure is the remainder of T less household and wage-earning labor, only equations for household and wage-earning labor are presented. Lastly, by of the construction of partner utility functions as weighted averages, the following identity holds:

$$
\begin{aligned}
\Theta_0 &= \alpha(1 - \lambda_1) + (1 - \alpha)(1 - \lambda_2) \qquad (1) \\
&= 1 - \alpha\lambda_1 - (1 - \alpha)\lambda_2
\end{aligned}
$$

Given this convention, the first order conditions for partner 1 are:

$$\frac{\partial W}{\partial h_1} = \frac{\delta_3 w_1}{M}\Theta_0 - \frac{\alpha\lambda_1}{l_1} \qquad (2)$$

$$\frac{\partial W}{\partial \tau_1} = \frac{\delta_1}{\tau_1}\Theta_0 - \frac{\alpha\lambda_1}{l_1} \qquad (3)$$

In effect, these equations reflect the efficiency with which an hour of either type of labor benefits the household good, less the foregone leisure. Increases to efficiency of income or household labor increase the marginal value of an hour spent on each respectively; however, there are diminishing marginal returns (as represented by the M and τ_1 in each equation's respective first denominators).

Solving the first order conditions for time use,[4] we find:

[4] In the interest of clarity, I do not yet include the cost function, which constrains the time use values. Actual equations used in estimation are discussed in Section 5.

$$h_1 = T(1-\Theta_1) - \frac{\Theta_1(Tw_2 + Y)}{w_1} \tag{4}$$
$$\text{where,} \quad \Theta_1 = \delta_1\Theta_0 + \alpha\lambda_1 = (1-\delta_1)\alpha\lambda_1 + \delta_1(1-(1-\alpha)\lambda_2)$$

$$\tau_1 = \Theta_2\left(T + \frac{Y + Tw_2}{w_1}\right) \tag{5}$$
$$\text{where,} \quad \Theta_2 = \delta_1\Theta_0 = -\delta_1\alpha\lambda_1 + \delta_1(1-(1-\alpha)\lambda_2)$$

In the optimal time use equations, the total preference for the household good is adjusted by the efficiency with which time spent towards it benefits the household. For wage-earning labor, this is used to determine the fraction of time not spent directly on the household or personal leisure, with an additional discount for the effect of wage on the household product. Similarly, household labor is a question of how efficient a partner is at producing the household good, how much the household benefits from the good, and a deflator for how impactful the partner's wage-earning labor is on the household good.

As would be expected, h_1 is decreasing in partner 2's wage (w_2), household non-labor income (Y), the opportunity cost of work (Θ_1), and the efficiency of partner 1's household labor (δ_1); increasing in each partner's leisure preference ($\lambda_{1,2}$); and either decreasing or increasing in the bargaining weight (α), conditional on whether partner 2's preference outweighs partner 1's or the reverse, respectively.

Similarly, τ_1 is increasing in non-labor income (Y), partner 2's wage (w_2), and the efficiency of partner 1's household labor (δ_1); decreasing in each partner's preference for leisure ($\lambda_{1,2}$); and either increasing or decreasing in the bargaining weight (α), conditional on whether partner 2's preference outweighs partner 1's or the reverse, respectively.

Imposed on these utilities are constraints on reasonable time use and efficiencies: total time use is set to 120 hours per week, negative time use is not permitted, and the household product is constructed to have linear returns to scale. Efficiency exponents are also presumed to be nonnegative. These constraints parallel those used by del Boca and Flinn, except that T is increased from 112 to account for the flexibility introduced by the cost function (see Section 5 for details).

Unlike the constrained Pareto optimality and endogenous household interaction models presented in del Boca and Flinn (2012), this model does not allow for noncooperative behavior within households, nor does it constrain household decisions to benefit each individual partner at least as much as the Nash equilibrium would. Further, this model does not capture households with more than two key decision-making partners, especially households

where the household product (household income and externalities) is significantly affected by other members. Moreover, this model does not account for households where wealth is not a public good—e.g. use of separate bank accounts.

5 Empirical Model

The model presented in the previous section is not yet viable for quantitative modeling. Impeding its use are six unobservable variables: $\alpha, \lambda_1, \lambda_2, \delta_1, \delta_2,$ and δ_3. Many of the variables in the behavioral model presented by del Boca and Flinn (2012) are not observed: partner preferences for leisure versus the household good and partner Cobb-Douglas exponents are of particular interest. A key motivation for the use of the del Boca and Flinn (2012) paper is their solution to this challenge: the method of simulated moments (McFadden, 1989). As in del Boca and Flinn (2012), we assume that α is 0.5 and that the household good has linear returns to scale ($\delta_3 = 1 - \delta_1 - \delta_2$). The former assumption reflects equal deference to each partner's preferences in the household decision. Thus, the parameters of interest relate to the distributions of $\lambda_1, \lambda_2, \delta_1,$ and δ_2. These shall be referred to as S_1. Because this sample is constructed entirely of different-sex households, men are all assigned to $j = 1$ and women to $j = 2$. This is to say, λ_1 and δ_1 reflect the preferences and Cobb-Douglas exponents (efficiencies) of men; λ_2 and δ_2 reflect the same for women. Following the approach of del Boca and Flinn (2012) to identify the distributions of these variables, we use the method of simulated moments (McFadden, 1989), estimating a 4×1 mean vector and a 4×4 covariance matrix (defined by 14 total parameters). The remainder of this section outlines the mathematical and computational steps involved in implementing this approach.

5.1 Method of Simulated Moments

Situated in the theoretical lineage of maximum likelihood estimators, the method of simulated moments identifies the parameters of an unobservable distribution, such that observable products of the distribution best match reality. Formally:

$$\theta_{MSM} = \text{argmin}_{\theta} \left(m - \hat{m}\left(\theta\right)\right)^T Q \left(m - \hat{m}\left(\theta\right)\right) \tag{6}$$

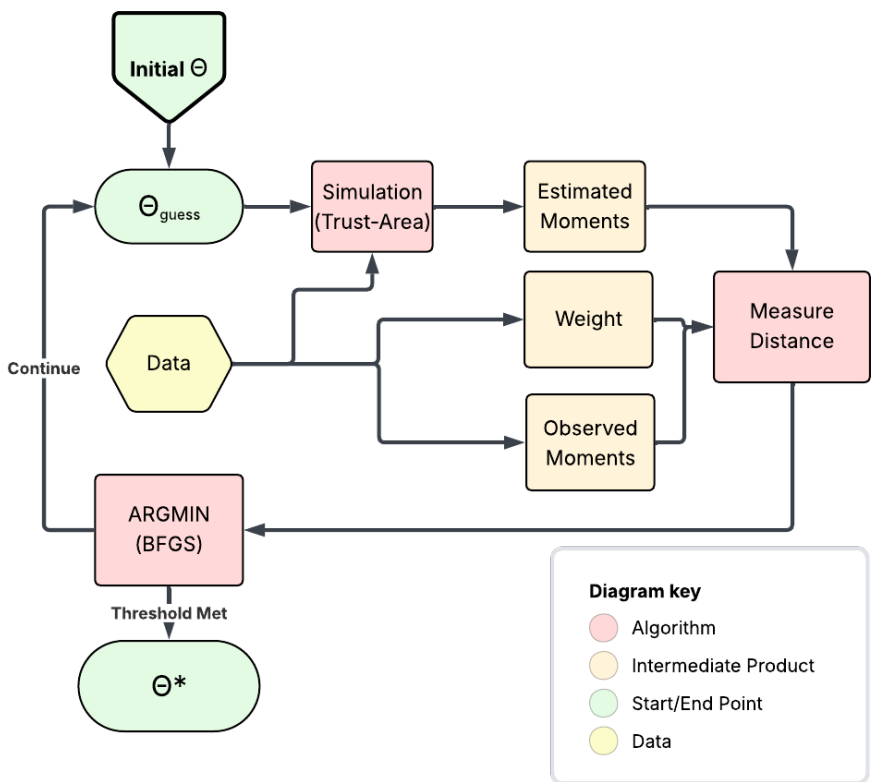
, where θ is a vector of the distribution parameters, m are the observed products (moments) of the distribution, $\hat{m}\left(\theta\right)$ are the estimated moments for a particular set of parameters, and Q is a weighting matrix. In the interest of extending the del Boca and Flinn (2012) analysis to new data, we use the same 21 moments presented in Table A.1 of their working

paper.[5] Q is defined using bootstrapped standard errors of the moments, as in del Boca and Flinn (2012). This is intended to de-emphasize noisy moments.

5.2 Implementation

Implementing the method of simulated moments involves: i) relating the unobserved distribution to observable statistics, ii) generating random draws from some parameter-defined distribution, iii) estimating observable statistics, iv) measuring the distance of estimates from actual observed statistics, and v) repeating ii-iv until the distance is sufficiently small. This process is described below, and the relationships between the algorithms and data used are visualized in Figure 2.

Figure 2: Flow Chart of MSM Procedure



The method of simulated moments (MSM) procedure is used to 'reverse engineer' unobservable parameters by matching observed moments, which are presumed to derive from the unobservable parameters (McFadden, 1989). Moments represent the means and standard deviations of (sometimes interacted) observed variables. A list of moments, as well as those reported by del Boca and Flinn (2012) may be found in Table 11.

[5] del Boca and Flinn (2012) state that they implement 23 moments, but their table of moments (reproduced here in Appendix C) only shows 21.

Operationalizing the household decision, we use the characterization presented in the Theoretical Model, with $\alpha, l_1, l_2,$ and δ_3 substituted based on the previously mentioned constraints: equal weight, total time use, and linear returns to scale. For notational clarity, the observable, exogenous income variables—$w_1, w_2,$ and Y—are referred to as S_2. Furthermore, the observable, endogenous time use variables—$h_1, h_2, \tau_1,$ and τ_2—are S_3. Additionally, to promote computational efficiency, a cost function is selected in lieu of constrained optimization:

$$\text{cost}\left(var\right) = e^{c(l-var)} + e^{c(var-u)}$$

, where c is some constant scaling factor (set to 10), l is the lower bound, and u is the upper bound. Thus, the implemented objective function was:

$$S_3^* = \text{argmin}_{S_3}\left(W\left(S_3\right) + \sum_{var \in S_3} \text{cost}\left(var\right)\right)$$

To derive estimates of S_1 from the estimated parameter vector Θ, we create a mean and covariance matrix, such that $\mu = \left[\Theta_1, \ldots, \Theta_4\right]^T$ and $\Sigma = C^T C$, where:

$$C = \begin{bmatrix} \exp(\theta_5) & \theta_6 & \theta_7 & \theta_8 \\ 0 & \exp(\theta_9) & \theta_{10} & \theta_{11} \\ 0 & 0 & \exp(\theta_{12}) & \theta_{13} \\ 0 & 0 & 0 & \exp(\theta_{14}) \end{bmatrix}$$

Derived from this, a 4×1 matrix, x, was drawn: $x \in^{i.i.d.} N\left(\mu, \Sigma\right)$. In turn, the following equations translate the infinite-domained random draws to the narrowly constrained values of unobserved variables in S_1:

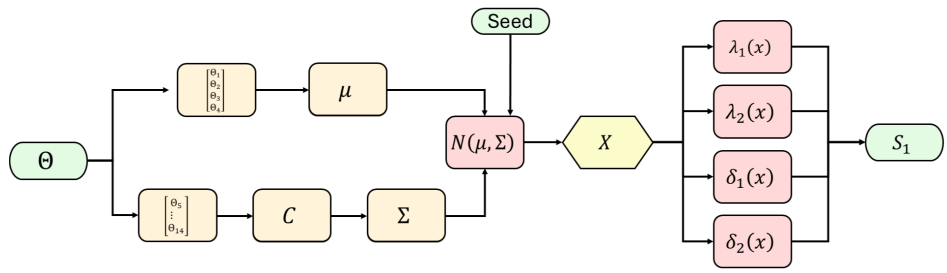
$$\begin{aligned} \lambda_1 &= \frac{1}{1 + e^{-x_1}} \\ \lambda_2 &= \frac{1}{1 + e^{-x_2}} \\ \delta_1 &= \frac{e^{x_3}}{1 + e^{x_3} + e^{x_4}} \\ \delta_2 &= \frac{e^{x_4}}{1 + e^{x_3} + e^{x_4}} \end{aligned} \tag{7}$$

These steps are visualized in Figure 3. Substituting a household's S_2 and a randomly drawn S_1 into the household objective function and optimizing for S_3 produces a dataset of estimated time use variables. This corresponds to the light red box labeled "Simulation (Trust-Area)" in Figure 2. Having thus generated a set of estimated time use variables, we join the household income variables and calculate the moments for this value of the parameter

vector. These moments are compared to the observed moments in the actual data, and the weighted differences are converted to a scalar cost value:

$$(m - \hat{m}(\theta))^T Q (m - \hat{m}(\theta))$$

Figure 3: Conversion of Theta to S_1



This flowchart describes the conversion from estimated parameters (Θ) to meaningful decision-making parameters (S_1). Random draws from a gaussian distribution characterized by Θ are mapped onto the range (0,1). Repeating this process produces a distribution of values for S_1.

The above steps describe testing one value of the parameter vector. The final step of estimation was to iterate, searching for some value of the parameter vector that would minimize the scalar cost value.

5.3 Bootstrapped S_1 Estimates

Because all of the parameters in the resulting Θ^* are transformed by some function in order to generate S_1, it is not possible to meaningfully interpret them. Therefore, we bootstrap estimates and standard errors of the distribution parameters of S_1. This process makes the strong assumptions of a parametric distribution of S_1 and independence between S_1 and S_2. For the initial replication, we simulated 1,000,000 draws from the distribution defined by Θ^*. Each draw was then converted to its corresponding values of S_1. This dataset was then resampled 10,000 times. The mean and standard deviation of each variable in S_1 was recorded. Means and standard errors of these distribution parameters were then calculated. All subsequent tests (robustness to sample, robustness to initial parameters, and estimates in novel years) used an identical procedure, except 1,000 draws were simulated and 100 resamplings were conducted. This was done for computational efficiency.

All estimations were run in MATLAB. The BFGS algorithm was used for estimating Θ^* and the trust-region algorithm (with a supplied gradient) was used to identify optimal household decisions. Data analysis and bootstrapped S_1 were completed in Python after all estima-

tions were obtained. The scripts for this project are available here: https://github.com/arietin/EconHonors

6 Results

The model as constructed converged with minimal exceptions.[6] Bootstrapped means and standard errors of point estimates are represented in Table 7, alongside the corresponding estimates from del Boca and Flinn (2012).

Likely due to the number of resamplings (10,000), standard errors are incredibly small and are reported in scientific notation. Because del Boca and Flinn (2012) do not report the number of times they resampled their data, an F-Test is not feasible. Sufficient information was present to determine whether the estimates were means of the distributions estimated by del Boca and Flinn (2012).

Differences were not statistically significant for five of the eight parameters. Significant differences are reported at the 95% confidence interval and above for $\mu(\lambda_1), \mu(\lambda_2)$, and $\sigma(\lambda_2)$.

The results of the three robustness checks— to sample, to initialization, and to time— are presented below.

6.1 Robustness to Sample

For each of the 25 resamplings, the model converged. Means and standard deviations of these samples are represented in Table 12. The estimated parameter vector $\hat{\Theta}_s^*$ for each of the 25 samples was subsequently used to generate 1000 S_1's. Bootstrapped average means and standard deviations of the preference and efficiency variables were calculated over 100 resamplings. The mean values and standard error of these 25 bootstrapped averages, alongside the previously reported results by del Boca and Flinn (2012), are reported in Table 8.

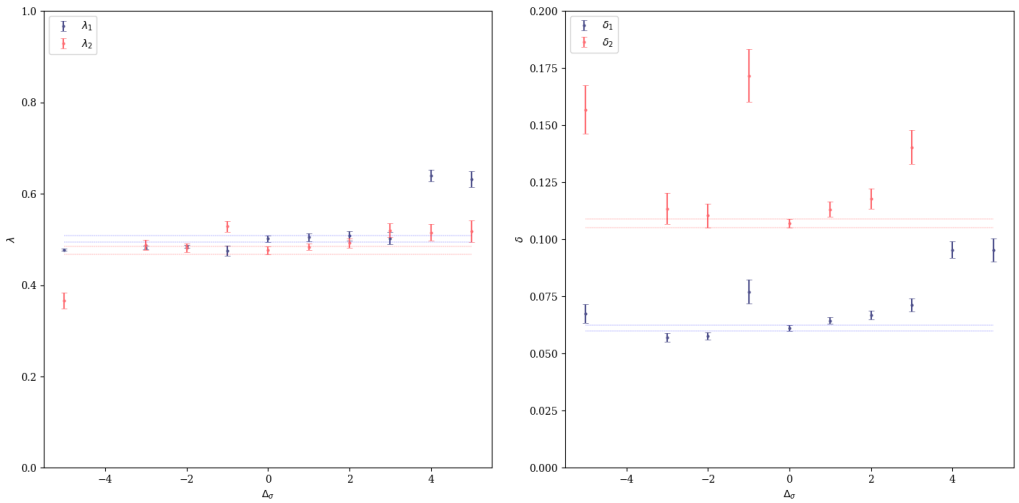
6.2 Robustness to Initial Parameters

Turning attention now from the choice of sample to the choice of initial conditions, the standard error of $\hat{\Theta}^*$ from Table 12 was used to generate 10 new initializations of Θ. Multiples between -5 and 5 of the standard error were added to the previously used initialization. All initializations except for $\hat{\Theta}^* - 4\sigma(\hat{\Theta}^*)$ converged.

Bootstrapped means and standard errors are represented in Figure 4. Exact estimates are also represented in Table 13, in Appendix D. Estimated $\hat{\Theta}^*$ for each initialization are reported in Table 16, located in Appendix D.

[6] Only when testing initialization at -4 standard deviations from the mean and 2023 data did the model fail to converge. See subsections 6.2 and 6.3.

Figure 4: Estimated Decision Parameters at Distance $\Delta_\sigma \sigma(\Theta_0)$ from Θ_0



(a) $\hat{\lambda}^*$ at varying Θ_0 (b) $\hat{\delta}^*$ at varying Θ_0

Estimates of parameters for men (subscript 1) and women (subscript 2), initialized at distance Δ_σ from Θ_0 used in replication. Lambdas reflect preference for personal time, deltas are Cobb-Douglas exponents, analogous to efficiency. The simulation $\Delta_\sigma = -4$ did not converge. Bootstrapped means and standard errors (n = 100 resamplings of d = 1000 draws from each $\hat{\Theta}_\Delta$). Error bars are at $2 \times \sigma(\mu(\text{estimate}))$.

6.3 Robustness over time

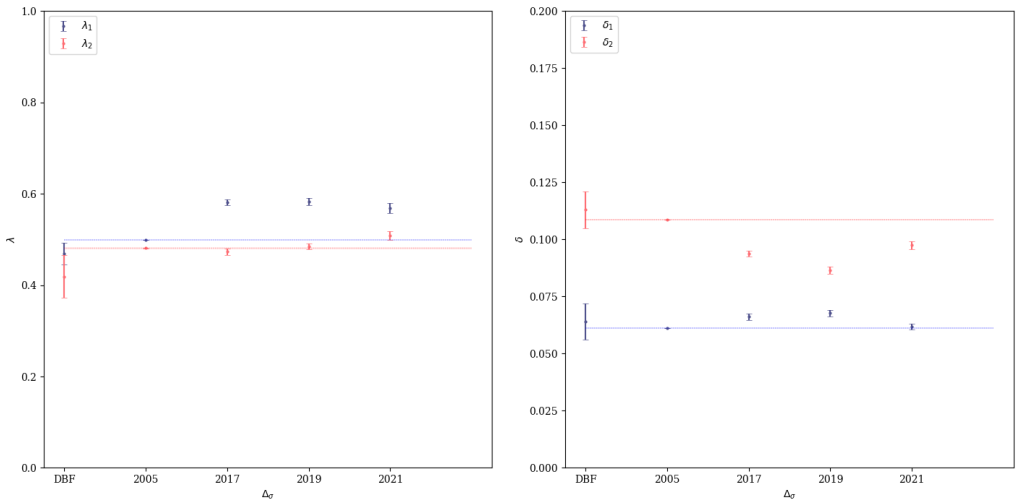
Testing the final question of continuity over time, we estimated $\hat{\Theta}^*$ using data from 2017, 2019, 2021, and 2023. Only estimates for 2023 failed to converge. The remaining estimates are reported in Table 15. Bootstrapped means and standard deviations of preference and efficiency parameters are shown in Figure 5. Exact estimates are expressed in Appendix D, Table 14.

7 Discussion

The focus of the present research has been to explore unobservable household time-use decision parameters, extending del Boca and Flinn's 2012 analysis. Novel tests of the method's robustness to sample selection, initial conditions, and samples from recent years have been presented. Replication was largely successful and each of the three tests of robustness—to sample, initialization, and year—produced meaningful results.

Analyses of the income and time-use data used in this replication show that the 2005 sample did not differ significantly from del Boca and Flinn's sample (see Table 4). Despite differences in data cleaning, only three of the eight estimated preference and efficiency parameters differed significantly from del Boca and Flinn's point estimates. Of the three sig-

Figure 5: Estimated Decision Parameters for Novel Years



(a) $\hat{\lambda}^*$ from different-year samples (b) $\hat{\delta}^*$ from different-year samples

Novel year estimates of parameters for men (subscript 1) and women (subscript 2). Lambdas reflect preference for personal time, deltas are Cobb-Douglas exponents, analogous to efficiency. The simulation for year 2023 did not converge. Bootstrapped means and standard errors (n = 100 resamplings of d = 1000 draws from each $\hat{\Theta}_{\Delta}$). Error bars are at $2 \times \sigma(\mu(\text{estimate}))$.

nificantly different estimates—$\mu(\lambda_1)$, $\mu(\lambda_2)$, and $\sigma(\lambda_2)$— one ($\mu(\lambda_1)$) remained below del Boca and Flinn's non-parametric estimate, though still greater than their parametric estimate.[7] Moreover, ordinality of estimates was maintained: $\mu(\lambda_1) > \mu(\lambda_2)$ and $\sigma(\lambda_1) < \sigma(\lambda_2)$ in both the present analysis and del Boca and Flinn's. Because the total amount of time available in the week was increased from 112 hours to 120 hours to account for the unconstrained optimization, via a cost function, of household decisions, the increased preference for leisure could reflect a desire by both men and women to have more than 6:50 of sleep per night. Further testing is needed to evaluate this potential explanation.

The initial replication was promising: differences in estimates were minimal and largely insignificant. It is entirely possible that these differences may be attributed to disparities in the samples, so having been convinced of sufficiently similar replication results, further analyses were warranted.

Testing the extent to which variations in sample affected results, analyses with resampled data again suggested that differences in sample resulted in minimal variation in point estimates, but that del Boca and Flinn's parameters of women's leisure preferences differed significantly from the presently estimated ones. Women in the present sample appear sig-

[7]Del Boca and Flinn estimate preference and efficiency parameters both parametrically and non-parametrically. To facilitate direct comparison, estimates in this thesis are derived parametrically.

nificantly more inclined to leisure and more homogenous in their inclination than del Boca and Flinn estimated. Therefore, estimates are concluded to be somewhat sample dependent. Future directions of research should explore the effect of relaxing or changing the selection criteria for the sample (e.g. the age range of partners) on estimates.

Testing the extent to which variations in initial conditions affected results, rerun analyses with new initial parameters showed broad stability in men's and women's preferences for leisure. However, efficiency estimates showed greater sensitivity to initial conditions. Because the initial conditions used in this analysis were selected to approximate del Boca and Flinn's best estimates, it is possible that this choice considerably affected the similarity of results: estimates seem sensitive to their initialization. Additional research might compare results using different search algorithms (e.g. trust-region, BFGS, L-BFGS, genetic algorithms, etc.) or more sensitive optimization tolerances.

Testing the continuity of estimates over time, analyses with data from 2017, 2019, and 2021[8] showed divergence in both preferences and efficiencies in the novel years. Men's preferences for leisure were elevated in all years tested, but women only showed elevated preferences for leisure in 2021, though it did not exceed men's estimated preference for leisure in that year. Estimated efficiency parameters, by contrast, were closer in each tested year than in 2005: both men and women's efficiencies were above and below, respectively, their 2005 estimates in 2017 and 2019. Only in 2021 did estimates show no difference in men's efficiency from 2005 data, but women still demonstrated lower efficiency as in 2017 and 2019. This novel year dataset is limited to only three observations and further analysis is warranted before drawing conclusions about preferences and efficiencies over time. The relationship between inequality in these estimates and other measures of gender inequality is a potential avenue for future research.

The exclusion of same-sex households in the present analyses should be remedied in subsequent literature. The demographic differences and lack of inclusion in the del Boca and Flinn analysis were used to justify their exclusion in these analyses. However, differences in time-use may be observed even when men and women in same-sex households have the same preferences and efficiencies as their different-sex household counterparts. In such cases, time use differences may result from variance in other exogenous conditions.[9]

Just as the full PSID sample contains far more varied types of households than were used, actual time-use decision dynamics are more nuanced than the pareto optimal behavioral regime used here. Del Boca and Flinn test alternative theoretical models, including

[8]The analysis of 2023 data did not converge and thus conclusions are not drawn from it.

[9]Exogenous to this analysis; mean number of dependents, wage, education, etc. may be endogenous in other models.

endogenous household interaction, where a household might choose to find a pareto-optimal solution or settle for the Nash equilibrium. This behavioral regime outperformed pareto-optimality on most measures. All variations of the theoretical model must balance computational viability with the description of real dynamics. Some additional nuances that might be justified are the use of a variable α term[10] (deference to partner one's preferences), allowing for multiple household goods,[11] or split bank accounts (so not all wages go to public good production).[12]

Moreover, this thesis primarily explored measures of internal validity: subsequent work might analyze the predictive power of these estimates in the context of policy analysis. Because a primary motivation for the use of the method of simulated moments is to conduct forward-looking policy analyses, confirming that MSM estimates (whether from this theoretical model or another) appropriately predict real behavior would allow for extensive experimentation in labor-market policy.

8 Conclusion

Traditional econometric tests of qualitative theoretical models provide limited opportunity for 'what if' policy analyses, being constrained to past events. Quantitative models of labor supply in an economy where households make joint decisions are therefore valuable in any number of government and private sector policy analyses. Prior research uses 2005 PSID data to derive the necessary unobserved parameters for quantitative modeling (Del Boca and Flinn, 2012). This research seeks to update estimates to align with contemporary data, now 15–20 years newer than the original del Boca and Flinn data. Replication was largely successful, and internal validity was tested, using different samples, starting conditions, and years. Thus, the first volley of theoretical, empirical, and computational frameworks for exploring present-day household time-use decisions has been fired.

[10] In different-sex households, this would be deference to the man. However, in same-sex households, determining who to label as 'partner 1' poses a challenge: using wage, age, or primary respondent status produce bimodal distributions of within-household disparities, meaning that means of intra-household differences—like α—often suggest incredible equality. Presuming a bimodal distribution of α might be a potential solution.

[11] PSID data now include more specific time-use options, so children might be their own good, as in Chiappori et al., 2024.

[12] Allowing for a private good as a Cobb-Douglas product of leisure and private income would raise the question of who determines the split of wages between private and public consumption—the earner or the household.

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Table 6: Summary Statistics by Gender, Novel Years

	2017 Gender		2019 Gender		2021 Gender		2023 Gender	
	Male	Female	Male	Female	Male	Female	Male	Female
N	1,106 (50.0%)	1,108 (50.0%)	1,126 (49.8%)	1,136 (50.2%)	1,046 (49.6%)	1,064 (50.4%)	1,044 (49.4%)	1,068 (50.6%)
Age	39.477 (4.864)***	38.161 (4.833)	39.724 (4.948)***	38.410 (4.871)	39.956 (4.857)***	38.552 (4.709)	40.405 (4.826)***	39.008 (4.709)
Years of completed education, imputed	14.262 (2.320)***	14.830 (2.185)	14.369 (2.267)***	14.986 (2.037)	14.543 (2.220)***	15.024 (2.075)	14.471 (2.258)***	15.113 (1.991)
Family unit size	3.892 (1.264)	3.894 (1.264)	3.875 (1.241)	3.864 (1.246)	3.931 (1.223)	3.910 (1.227)	3.903 (1.264)	3.881 (1.264)
Total family income last year	137.680 (87.937)	137.352 (87.896)	145.236 (107.781)	144.760 (107.485)	153.844 (98.476)	152.502 (97.760)	158.373 (127.678)	156.860 (125.871)
Prior year family labor earnings	131.954 (87.187)	131.592 (87.145)	139.520 (107.451)	139.044 (107.171)	146.219 (98.162)	144.997 (97.408)	152.432 (126.583)	150.969 (124.923)
Prior year labor earnings	79.268 (65.283)***	52.529 (44.028)	84.148 (87.673)***	55.261 (44.128)	87.374 (77.342)***	58.475 (47.276)	88.660 (83.833)***	63.320 (68.473)
Prior year household weekly non-labor income	110.117 (198.330)	110.762 (198.373)	109.931 (204.701)	109.916 (204.728)	146.640 (222.936)	144.324 (221.197)	114.265 (221.174)	113.289 (218.655)
Prior year hourly wage	25.656 (21.082)***	21.404 (27.562)	27.024 (26.319)***	21.391 (15.900)	29.501 (25.279)***	23.923 (21.453)	29.317 (25.812)***	24.505 (24.778)
Race							*	
White	0.696	0.711	0.702	0.717	0.711	0.722	0.716	0.713
Black	0.227	0.200	0.226	0.205	0.207	0.183	0.215	0.187
Other	0.074	0.084	0.069	0.073	0.077	0.089	0.068	0.095
Unknown	0.003	0.005	0.003	0.004	0.005	0.006	0.002	0.005
College degree	***		***		***		***	
No	0.451	0.314	0.446	0.316	0.421	0.314	0.433	0.314
Yes	0.460	0.592	0.485	0.610	0.511	0.616	0.495	0.625
Unknown	0.089	0.094	0.069	0.074	0.069	0.070	0.072	0.061
Currently working	*		***		**		***	
0	0.040	0.056	0.027	0.062	0.044	0.069	0.033	0.060
1	0.960	0.944	0.973	0.938	0.956	0.931	0.967	0.940
Full-time status last year	***		***		***		***	
Part-time (50-1749 hours)	0.100	0.348	0.091	0.332	0.153	0.344	0.108	0.284
Full-time (1750+ hours)	0.900	0.652	0.909	0.668	0.847	0.656	0.892	0.716
Low-income family last year								
0	0.979	0.979	0.991	0.991	0.993	0.993	0.989	0.989
1	0.021	0.021	0.009	0.009	0.007	0.007	0.011	0.011

Demographics of samples used for estimation in novel years. Standard deviations in parentheses for continuous variables. Significance scores are noted on the male entry for a given year/statistic combination. Significance scores based on p-value for testing the equality between genders within a year, using t-test for continuous variables and Pearson χ^2 tests for factor variables. Total family income, total labor earnings, and individual labor earnings are in \$1000's, all prices in 2021 USD. Key: * = 90% confidence, ** = 95% confidence, *** = 99% confidence.

Table 7: Bootstrapped Mean and S.E. of Parametric Point Estimates

	Present Estimates		Del Boca and Flinn, 2012	
	Estimate	S.E.	Estimate	S.E.
$\mu(\lambda_1)$	0.499**	1.1e-04	0.469	0.012
$\sigma(\lambda_1)$	0.113	7.3e-05	0.152	0.033
$\mu(\lambda_2)$	0.481***	1.3e-04	0.419	0.023
$\sigma(\lambda_2)$	0.132***	8.3e-05	0.233	0.022
$\mu(\delta_1)$	0.061	1.8e-05	0.064	0.004
$\sigma(\delta_1)$	0.018	1.5e-05	0.033	0.011
$\mu(\delta_2)$	0.109	3.4e-05	0.113	0.004
$\sigma(\delta_2)$	0.034	3.0e-05	0.029	0.008

Bootstrapped distribution characteristics of variables in S_1, estimated using 2005 data. Values presented alongside estimates from del Boca and Flinn (2012). Due to the large number of resamplings, bootstrapped standard errors were extremely low. Therefore, scientific notation were used. Null Hypothesis: the mean of the distribution characterized by del Boca and Flinn's estimates is that which is presently estimated. Key: * = 90% confidence, ** = 95% confidence, *** = 99% confidence.

Table 8: Sample-Robustness of Unobserved Preference and Efficiency Estimates

	Sample-Robust		Del Boca and Flinn, 2012	
Characteristic	Estimate	S.E.	Estimate	S.E.
$\mu(\lambda_1)$	0.501	0.020	0.469	0.012
$\sigma(\lambda_1)$	0.125	0.030	0.152	0.033
$\mu(\lambda_2)$	0.468	0.015	0.419***	0.023
$\sigma(\lambda_2)$	0.123	0.040	0.233***	0.022
$\mu(\delta_1)$	0.061	0.004	0.064	0.004
$\sigma(\delta_1)$	0.022	0.010	0.033	0.011
$\mu(\delta_2)$	0.109	0.009	0.113	0.004
$\sigma(\delta_2)$	0.045	0.022	0.029	0.008

Distribution characteristics of variables in S_1, estimated using 25 resamplings 2005 data. Null Hypothesis: means of the presently estimated characteristics are identical to del Boca and Flinn's. Key: * = 90% confidence, ** = 95% confidence, *** = 99% confidence.

A Key PSID Variables

Table 9: Key PSID Variables and Meanings

Variable	Meaning
adhrs_house	Absolute Difference: time in housework
adhrs_pers	Absolute Difference: time in personal care
adhrs_shop	Absolute Difference: time in shopping
adhrs_chld	Absolute Difference: time in child care
adhrs_care	Absolute Difference: time in adult care
adhrs_volu	Absolute Difference: time in volunteering
adhrs_educ	Absolute Difference: time in education
adhrs_leis	Absolute Difference: time in leisure
adhrs_work	Absolute Difference: time spent working
intdate	Date of Interview
idind	Person ID
famid	Family ID
famid1968	1968 FAMILY IDENTIFIER
famid1968b	Original family ID 1968
year	Year
seqnum	Sequence number
seqcat	In family at survey time seqcat
relation	Relation to head of household
relcat	Relation to household head, categories relcat
whoresp	Indicates the specific respondent answering the questions
respond	Relationship Code from question ER85379L
marpair	Married pairs/couples in family marpair
spouse	Head-spouse distinction spouse
sample	PSID samples sample
lweight	Individual longitudinal weight
cweight	Individual cross-sectional weight, 1997-
statefips	FIPS of state residence statefips
statechild	State FIPS where the respondent grew up
female	Female
sexfam	Sex from family files ER82019L
age_orig	Age in original individual data
age	Age
agefam	Age from family files
race	Race race
eduyrs	Years of completed education, imputed
college	College degree college
edufath	Education of father edupar
edumoth	Education of mother edupar
marstat	Marital status marstat
famsize	Family unit size ER82017L
nbirths	# Live births to this individual nbirths
nchildren	Number of children in family unit (ages 0-17) ER82022L
famincly	Total family income last year (2021 prices)
lowincly	Low-income family last year
famexp	Total family expenditure (2021 prices)
empstat	Employment status, 1979- empstat_new
employdB	Currently working
exper	Years of experience since age 18 ER85166L
hoursly	Annual work hours last year
labincly	Annual labor earnings last year (2021 prices)
wagely	Hourly wage last year (2021 prices)
wagemain	Hourly wage rate at main job (2021 prices)
hltstat	Health status ER84520L
hrs_house	Time in housework
hrs_pers	Time in personal care
hrs_shop	Time in shopping
hrs_chld	Time in child care
hrs_care	Time in adult care
hrs_volu	Time in volunteering
hrs_educ	Time in education
hrs_leis	Time in leisure
hrs_work	Time spent working
lcpi	Consumer price index (Base year 2021[June-June])
cpi	Consumer price index (Base year 2021[June-June])
hhintdate	Year-month of family interview
married	Married or living together
bplace	Place of growing up bplace
onleave	Currently on temporary leave (e.g., sick/maternity)
unempld	Currently unemployed
housekp	Currently keeping house
fulltimely	Full-time status last year fulltimely
hlthbad	Poor health
samesex	Same sex couple samesexl
kfamincly	Total family income last year (2021 prices) in $1000s
klabincly	Annual labor earnings last year (2021 prices) in $1000s

Variables and their labels used in stata analyses.

B PSID Questions

Table 10: Sample PSID Time-Use Questions

Variable	Codebook
hrs_house	F1a. These next few questions are about how (you spend/[Reference Person] spends) (your/his/her) time (CURRENTLY WORKING: when [you are/[he/she] is] not working). In a typical week, how many hours (do you/does [he/she]) spend doing housework, for example, cooking, cleaning, and other work around the house?
hrs_pers	F1b. (In a typical week, how many hours [do you/does [he/she]] spend) Doing personal care activities, for example, grooming, getting ready for the day, or taking care of (your/his/her) health needs?
hrs_shop	F1c. (In a typical week, how many hours [do you/does [he/she]] spend) Shopping, for example, buying groceries or clothes, or shopping online?
hrs_child	F1d. (In a typical week, how many hours [do you/does [he/she]] spend) Caring for or looking after children? (CURRENTLY WORKING: Exclude hours providing care if this is [your/his/her] job.)
hrs_care	F1d2. (In a typical week, how many hours [do you/does [he/she]] spend) Caring for or looking after an adult who needs assistance with daily activities? (CURRENTLY WORKING: Exclude hours providing care if this is [your/his/her] job.)
hrs_volu	F1e. (In a typical week, how many hours [do you/does [he/she]] spend) Volunteering, for example, at religious organizations, schools, hospitals, charities, or community organizations?
hrs_educ	F1f. (In a typical week, how many hours [do you/does [he/she]] spend) Doing educational activities, for example, taking classes, doing homework, or studying for exams? (>1 FU MEMBER: Exclude time spent helping others with homework.)
hrs_lesi	F1g. (In a typical week, how many hours [do you/does [he/she]] spend) Doing leisure activities for enjoyment, for example, watching TV, doing physical activities that (you enjoy/[he/she] enjoys), going online, or spending time with friends?
hrs_work	BC60a. These next questions are about the time (you spend/Reference Person spends) working in a typical week. In a typical week, how many hours (do you/does [he/she]) spend working for pay?–REFERENCE PERSON

Sample text of time use questions and associated variables as seen in the PSID codebook.

C Del Boca and Flinn's Moments Table

Table 11: Moments as reported in Del Boca and Flinn, 2012

Sample Characteristic	**Sample Value**
Average h_1	45.706
Average h_2	38.588
Average τ_1	7.787
Average τ_2	14.92
St. Dev. h_1	8.546
St. Dev. h_2	10.512
Average ($h_1 \times h_2$)	1772.074
Average ($h_1 \times$ Y)	5530.755
Average ($h_2 \times$ Y)	4580.739
Average ($h_1 \times w_1$)	1009.293
Average ($h_2 \times w_2$)	618.773
Average ($\tau_1 \times$ Y)	947.62
Average ($\tau_2 \times$ Y)	1876.535
St. Dev. τ_1	6.418
St. Dev. τ_2	9.423
Average ($h_1 \times w_2$)	722.88
Average ($h_2 \times w_1$)	830.367
Average ($h_1 \geq 40$)	0.955
Average ($h_2 \geq 40$)	0.694
Average ($25 \leq h_1 < 40$)	0.0346
Average ($25 \leq h_2 < 40$)	0.21

Observed moments in Del Boca and Flinn, 2012. These are listed in their prior working paper as Table A.1 on page 41 (Flinn and Daniela Del Boca, 2009).

D Supplemental Tables

Table 12: Sample-Robust 2005 Parameters

	Mean	STD
Θ_1	0.007	0.087
Θ_2	-0.132	0.072
Θ_3	-2.684	0.065
Θ_4	-2.119	0.093
Θ_5	-0.654	0.257
Θ_6	-0.263	0.321
Θ_7	0.178	0.170
Θ_8	-0.100	0.172
Θ_9	-0.981	0.266
Θ_{10}	-0.111	0.198
Θ_{11}	-0.077	0.290
Θ_{12}	-1.438	0.051
Θ_{13}	-0.006	0.255
Θ_{14}	-1.359	0.081

Means and standard deviations for each parameter of Θ, as estimated 25 times, each with a different seed dictating the resampled (with replacement) dataset. Seeds were set to be integers 0–24, inclusive.

Table 13: Unobserved Preference and Efficiency Estimates, by Starting Distance in SD from 2005 Sample-Robust Θ_0

	-5.000000	-3.000000	-2.000000	-1.000000	0.000000	1.000000	2.000000	3.000000	4.000000	5.000000
$\mu(\lambda_1)$	0.477	0.482	0.485	0.475	0.502	0.505	0.508	0.503	0.639	0.631
	(0.001)	(0.002)	(0.002)	(0.006)	(0.003)	(0.005)	(0.005)	(0.006)	(0.006)	(0.009)
$\sigma(\lambda_1)$	0.029	0.061	0.062	0.184	0.115	0.135	0.151	0.195	0.217	0.257
	(0.001)	(0.001)	(0.001)	(0.003)	(0.003)	(0.003)	(0.003)	(0.004)	(0.004)	(0.004)
$\mu(\lambda_2)$	0.366	0.488	0.482	0.528	0.476	0.484	0.492	0.520	0.515	0.518
	(0.009)	(0.006)	(0.005)	(0.006)	(0.004)	(0.004)	(0.005)	(0.007)	(0.009)	(0.012)
$\sigma(\lambda_2)$	0.272	0.168	0.161	0.208	0.133	0.129	0.173	0.235	0.295	0.329
	(0.005)	(0.003)	(0.003)	(0.003)	(0.003)	(0.002)	(0.003)	(0.004)	(0.004)	(0.004)
$\mu(\delta_1)$	0.068	0.057	0.058	0.077	0.061	0.064	0.067	0.071	0.095	0.095
	(0.002)	(0.001)	(0.001)	(0.003)	(0.001)	(0.001)	(0.001)	(0.001)	(0.002)	(0.002)
$\sigma(\delta_1)$	0.072	0.026	0.025	0.079	0.018	0.024	0.030	0.042	0.060	0.070
	(0.003)	(0.001)	(0.001)	(0.004)	(0.0)	(0.001)	(0.001)	(0.001)	(0.002)	(0.002)
$\mu(\delta_2)$	0.157	0.113	0.110	0.172	0.107	0.113	0.118	0.140	0.224	0.247
	(0.005)	(0.003)	(0.003)	(0.006)	(0.001)	(0.002)	(0.002)	(0.004)	(0.007)	(0.008)
$\sigma(\delta_2)$	0.210	0.110	0.078	0.202	0.034	0.055	0.074	0.110	0.211	0.253
	(0.006)	(0.004)	(0.003)	(0.006)	(0.001)	(0.002)	(0.003)	(0.004)	(0.005)	(0.006)

Distribution characteristics of variables in S_1, with estimation being initialized at 11 distances from the original guess used in replication. The simulation $\Delta_\sigma = -4$ did not converge. Bootstrapped means and standard errors are reported. The vector added or subtracted from the mean was the standard deviation identified in the 25-sample sample robustness test (see Section 6.1).

Table 14: Bootstrapped Point Estimates and Standard Errors for Novel Years

	del Boca and Flinn		2005		2017		2019		2021	
	Estimate	S.E.	Estimate	S.E.	Estimate	S.E.	Estimate	S.E.	Estimate	S.E.
$\mu(\lambda_1)$	0.469	0.012	0.499	1.1e-04	0.581	0.003	0.583	0.004	0.569	0.005
$\sigma(\lambda_1)$	0.152	0.033	0.113	7.3e-05	0.116	0.002	0.119	0.002	0.161	0.003
$\mu(\lambda_2)$	0.419	0.023	0.481	1.3e-04	0.473	0.003	0.485	0.003	0.508	0.005
$\sigma(\lambda_2)$	0.233	0.022	0.132	8.3e-05	0.111	0.002	0.111	0.002	0.152	0.003
$\mu(\delta_1)$	0.064	0.004	0.061	1.8e-05	0.066	0.001	0.068	0.001	0.062	0.001
$\sigma(\delta_1)$	0.033	0.011	0.018	1.5e-05	0.020	0.001	0.021	0.001	0.022	0.001
$\mu(\delta_2)$	0.113	0.004	0.109	3.4e-05	0.094	0.001	0.086	0.001	0.097	0.001
$\sigma(\delta_2)$	0.029	0.008	0.034	3.0e-05	0.024	0.001	0.023	0.001	0.029	0.001

Bootstrapped means and standard errors are reported alongside those by del Boca and Flinn (2012). Low SE in 2005 reflect the relatively larger number of draws and resamplings (1,000,000 vs 1,000 and 10,000 vs 100, respectively).

Table 15: Estimated Theta Values for Novel Years

	2005	2017	2019	2021
Θ_1	0.007	0.326	0.336	0.353
Θ_2	-0.132	-0.069	-0.069	0.084
Θ_3	-2.684	-2.609	-2.587	-2.672
Θ_4	-2.119	-2.235	-2.310	-2.186
Θ_5	-0.654	-0.677	-0.676	-0.308
Θ_6	-0.263	-0.274	-0.284	0.553
Θ_7	0.178	0.178	0.181	0.247
Θ_8	-0.100	-0.097	-0.089	0.131
Θ_9	-0.981	-0.982	-0.985	-0.948
Θ_{10}	-0.111	-0.134	-0.128	-0.186
Θ_{11}	-0.077	-0.053	-0.047	-0.098
Θ_{12}	-1.438	-1.438	-1.434	-1.505
Θ_{13}	-0.006	-0.008	-0.001	0.183
Θ_{14}	-1.359	-1.362	-1.359	-1.420

Estimated Θ^* for years 2005, 2017, 2019, and 2021. Estimation for 2023 did not converge. 2005 estimates are those estimated here, used as starting point for MSM estimation in novel years.

Table 16: Unobserved Parameter Estimates, by Starting Distance in SD from 2005 Bootstrap Θ_0

	Θ_1	Θ_2	Θ_3	Θ_4	Θ_5	Θ_6	Θ_7	Θ_8	Θ_9	Θ_{10}	Θ_{11}	Θ_{12}	Θ_{13}	Θ_{14}
-5	-0.102	-0.634	-2.711	-2.507	-2.178	-1.592	-0.840	-0.857	-2.248	-0.983	-1.449	-1.693	-1.275	-1.762
-4	DNC	DNC	DNC	DNC	DNC	DNC	DNC	DNC	DNC	DNC	DNC	DNC	DNC	DNC
-3	-0.072	-0.066	-2.759	-2.354	-1.408	-0.717	-0.382	-0.598	-1.616	-0.291	-0.558	-1.663	-0.651	-1.625
-2	-0.054	-0.092	-2.750	-2.311	-1.400	-0.656	-0.280	-0.329	-1.459	-0.305	-0.537	-1.547	-0.477	-1.522
-1	-0.067	0.125	-2.611	-2.259	-0.154	-0.183	0.764	-0.719	-0.056	0.789	1.569	-1.739	0.267	-2.160
0	-0.003	-0.082	-2.645	-2.081	-0.740	-0.324	0.141	-0.139	-0.761	0.134	0.193	-1.434	0.052	-1.367
1	0.013	-0.073	-2.616	-2.088	-0.532	-0.088	0.282	-0.068	-0.585	0.250	0.384	-1.424	0.292	-1.344
2	0.004	-0.008	-2.585	-2.080	-0.429	-0.153	0.358	0.006	-0.275	0.343	0.558	-1.398	0.466	-1.309
3	0.015	0.126	-2.546	-2.016	-0.123	-0.274	0.560	0.140	0.140	0.403	0.769	-1.401	0.594	-1.431
4	0.672	0.007	-2.131	-1.672	0.130	1.283	0.681	0.690	0.142	0.791	1.162	-1.228	1.019	-1.036
5	0.759	0.079	-2.066	-1.579	0.387	1.604	0.851	0.862	0.407	0.989	1.452	-1.176	1.274	-0.955
σ from Table 12	0.087	0.072	0.065	0.093	0.257	0.321	0.170	0.172	0.266	0.198	0.290	0.051	0.255	0.081

Estimated values for each parameter in Θ, with estimation being initialized at 11 distances from the original guess used in replication. The simulation starting at $\theta_{0,2005} - 4\sigma(\theta^*_{2005})$ did not converge. STD from table 12 included for reference.